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Knowledgeless Language Models

Anonymizing Named Entities at Pretraining Time Suppresses Parametric Factual Recall While Improving Contextual QA, Fact Verification, and Hallucination Detection

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Language models encode factual knowledge in their parameters during pretraining, and this parametric knowledge is a double-edged sword. It enables impressive zero-shot question answering but also produces confident hallucinations when the model’s memorized facts are outdated, incomplete, or contradicted by the provided context. KLLMs (Knowledgeless Language Models) address this at the source: instead of correcting parametric recall through fine-tuning or prompting, they prevent it from forming in the first place by anonymizing named entities before pretraining begins.

AT A GLANCE

Problem: LLMs memorize entity-linked facts during pretraining and hallucinate them confidently when context contradicts or omits them.

Why it matters: Hallucination and parametric override are root causes of LLM unreliability in knowledge-intensive tasks; existing mitigations act at fine-tuning or inference time.

Method: Named entities in the pretraining corpus are replaced with consistent-within-document, random-across-documents placeholder tokens, blocking entity-linked factual supervision.

Result: +12.2% relative on open-book NaturalQuestions; +8.5% on FEVER; +9.3% on HaluEval; near-zero perplexity cost.

Evidence: 125M, 350M, and 1.3B model scales; five benchmarks spanning QA, fact verification, and hallucination detection.

The Big Picture

When a language model is asked “Who directed Oppenheimer?” without any context, it answers from its parameters. When it is asked the same question with a context passage, it should answer from that passage — but it often

does not, especially when the parametric answer conflicts with the contextual one.

This *parametric override* problem is distinct from the out-of-distribution problem: it occurs even when the model has seen the correct answer in context. The issue is that parametric knowledge has been learned so strongly during pretraining that it dominates contextual signals in downstream deployment.

KLLMs test a simple hypothesis: if named entities are removed from pretraining data (replaced by anonymized placeholders), the model cannot form entity-specific parametric associations in the first place, making it structurally dependent on context for entity-linked facts.

Why Existing Approaches Fall Short

Retrieval-augmented generation (RAG) mitigates hallucination at inference time but does not change the model’s internal representation or its tendency to override retrieved context with parametric beliefs. Fine-tuning with contrastive or knowledge-editing objectives can suppress specific facts but does not address the pretraining-level memorization mechanism.

Prompting strategies (“base your answer only on the passage”) improve context use in controlled settings but fail in adversarial cases where the model’s high-confidence parametric recall outweighs the instruction. KLLMs target the fundamental source: the pretraining signal that creates entity-linked parametric knowledge.

Core Mechanism

A named entity recognition (NER) model processes each document and identifies spans corresponding to persons, organizations, locations, dates, and named products. Each unique entity within a document is replaced with a type-annotated placeholder: [PERSON_42], [ORG_7], etc.

Critically, the mapping from entities to placeholders resets between documents. The same person gets placeholder [PERSON_42] in one document and [PERSON_19] in another. This prevents the model from learning cross-document entity associations — the mechanism by which parametric factual knowledge accumulates.

Within a document, coreference is preserved: if [PERSON_42] is the subject of the first sentence, it remains [PERSON_42] throughout, so the model still learns intra-document discourse structure. Linguistic syntax, semantic roles, and factual relations within each document are fully intact; only entity identity across documents is suppressed.

Technical Formulation

Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be the pretraining corpus. Let $E(d_i) = \{e_1, \dots, e_k\}$ be the set of unique entities in document d_i . The anonymized document is:

$$\tilde{d}_i = \sigma_i(d_i)$$

where σ_i is a randomly sampled bijection from entities in d_i to type-specific placeholder tokens, drawn fresh for each document. The KLLM is trained with the standard autoregressive cross-entropy loss on $\{\tilde{d}_i\}$; no special objective is needed.

The cross-document anonymization satisfies: for any two documents $d_i \neq d_j$ and any entity e , with probability 1 the placeholder for e in $\tilde{d}_i$ differs from its placeholder in $\tilde{d}_j$.

Experimental Findings

Benchmark	KLLM-1.3B	Baseline-1.3B
NQ open-book EM $\uparrow$	43.1%	38.4%
NQ closed-book EM $\uparrow$	6.2%	18.7%
FEVER accuracy $\uparrow$	83.7%	79.2%
HaluEval $\uparrow$	72.4%	65.1%
TruthfulQA $\uparrow$	48.3%	41.7%
Perplexity $\downarrow$	14.3	14.1

The closed-book NQ drop (18.7% $\rightarrow$ 6.2%) confirms that entity anonymization successfully suppresses parametric factual recall. The open-book NQ gain (+12.2% relative) confirms that the model compensates by better utilizing context. The perplexity difference of 0.2 points (1.4% relative) confirms that linguistic competence is essentially unaffected.

Ablations and Interpretation

Partial anonymization: Anonymizing only person entities (not organizations or locations) yields approximately half the HaluEval improvement, suggesting that all entity types contribute to the parametric recall problem.

Fixed vs. random cross-document mapping: Using a fixed entity-to-placeholder mapping across all documents partially re-introduces factual associations (the model can form cross-document entity statistics by token co-occurrence). The random-per-document approach is essential for full suppression.

Downstream fine-tuning: KLLMs fine-tuned on Wikipedia-based QA recover most of the closed-book performance while retaining the open-book advantage, suggesting that the intervention targets pretraining memorization specifically rather than the model’s capacity to learn facts at all.

Scale: The open-book improvement grows from +6.1% at 125M to +12.2% at 1.3B, suggesting that larger KLLMs

are better at leveraging context when parametric recall is suppressed.

Reference

Roi Cohen et al. **Knowledgeless Language Models: Suppressing Parametric Recall for Evidence-Grounded Language Modeling**. arXiv:2607.12831, July 2026. <https://arxiv.org/abs/2607.12831>